

Joint J-Strict Continuity of the Multiplier Jordan Product

Candidate full solution for the FA/Banach discovery run

23 July 2026

Source questions and answer

The source is Fernandez-Polo, Garces, Li, and Peralta, *On the strict topology of the multipliers of a JB*-algebra*, arXiv:2210.13353 [1]. On page 25 it asks:

(Q1) Is the Jordan product of $M(\mathfrak{A})$ separately J-strict continuous (on bounded sets)?

(Q2) Is the Jordan product of $M(\mathfrak{A})$ jointly J-strict continuous on bounded sets?

Both answers are affirmative. The source page is included as `source_paper.pdf`; the exact statement is reproduced below.

Let $\mathfrak{A}$ be a JB*-algebra. The following two natural questions remain open:

(Q1) Is the Jordan product of $M(\mathfrak{A})$ separately J-strict continuous (on bounded sets)?

(Q2) Is the Jordan product of $M(\mathfrak{A})$ jointly J-strict continuous on bounded sets?

A bounded search on 23 July 2026 used the arXiv id, exact title, exact question phrases, *Jordan product J-strict topology multipliers*, and the published DOI 10.1007/s13398-023-01476-w, including citation-oriented variants. It found the source and bibliographic mirrors but no later work claiming either answer. This is bounded negative evidence, not a comprehensive novelty guarantee.

Notation and standard facts

Let $\mathfrak{A}$ be a JB*-algebra and let $M(\mathfrak{A})$ be its unital multiplier JB*-algebra. Its J-strict topology is generated by

$$\rho_a(z) = \|z \circ a\|, \quad a \in \mathfrak{A}, \quad z \in M(\mathfrak{A}).$$

The source proves that J-strict bounded sets in $M(\mathfrak{A})$ are exactly the norm-bounded sets [1]. It also recalls that $\mathfrak{A}$ has a positive contractive approximate unit (e_γ) such that

$$\|U_{1-e_\gamma}(a)\| \longrightarrow 0 \quad (a \in \mathfrak{A}). \quad (1)$$

For elements of a Jordan algebra, write

$$U_x(t) = 2(x \circ t) \circ x - x^2 \circ t$$

and introduce the symmetric bilinearization

$$U_{x,y}(t) = (x \circ t) \circ y + (y \circ t) \circ x - (x \circ y) \circ t.$$

Thus $U_{x+y} = U_x + U_y + 2U_{x,y}$, and norm submultiplicativity gives

$$\|U_{x,y}(t)\| \leq 3\|x\|\|y\|\|t\|. \quad (2)$$

We use the fundamental identity

$$U_{U_x(y)} = U_x U_y U_x \quad (3)$$

and the standard JB inequality

$$\|a \circ b\|^2 \leq \|a\| \|U_b(a)\| \quad (a \geq 0, b = b^*). \quad (4)$$

These are precisely the two standard facts recalled in the source [1, 2].

Proof intuition

Fix a positive contractive approximate-unit element e , and put $c = \mathbf{1} - e$. Split a multiplier z into

$$z = L_e z + R_e z, \quad R_e z = U_c z.$$

The local part $L_e z$ belongs to $\mathfrak{A}$, and if z tends J-strictly to zero then $L_e z$ tends to zero in norm for this fixed e . The remaining tail is not norm-small, but products of two tails remain tails:

$$U_c x \circ U_c y = U_c(U_{x,y}(c^2)).$$

The JB inequality and the fundamental identity estimate this last expression against $\|U_c(a)\|$ for each positive test element $a \in \mathfrak{A}$. Equation (1) makes that quantity uniformly small on norm-bounded x, y . This is the Jordan substitute for moving an element of an associative ideal through a product.

Localization lemmas

Lemma 1 (local-tail decomposition). *Let $e \in \mathfrak{A}$ be a positive contraction and put $c = \mathbf{1} - e$. For every $z \in M(\mathfrak{A})$,*

$$R_e z := U_c z, \quad L_e z := z - U_c z = 2e \circ z - 2(e \circ z) \circ e + e^2 \circ z.$$

In particular $L_e z \in \mathfrak{A}$, and

$$\|L_e z\| \leq 4\rho_e(z) + \rho_{e^2}(z). \quad (5)$$

Proof. Since $c = \mathbf{1} - e$, direct expansion of the quadratic representation gives

$$U_c z = z - 2e \circ z + 2(e \circ z) \circ e - e^2 \circ z.$$

The multiplier property puts $e \circ z$ and $e^2 \circ z$ in $\mathfrak{A}$; hence $L_e z \in \mathfrak{A}$. The displayed norm estimate follows from $\|e\| \leq 1$ and norm submultiplicativity. $\square$

Lemma 2 (uniform quadratic-tail estimate). *Let $c = c^*$ be a contraction in $M(\mathfrak{A})$, $w = w^* \in M(\mathfrak{A})$, and $a \in \mathfrak{A}_+$. Then*

$$\|U_c w \circ a\|^2 \leq \|a\| \|w\|^2 \|U_c(a)\|. \quad (6)$$

Proof. Apply (4) with $b = U_c w$, then use (3):

$$\begin{aligned}\|U_c w \circ a\|^2 &\leq \|a\| \|U_{U_c w}(a)\| \\ &= \|a\| \|U_c U_w U_c(a)\| \\ &\leq \|a\| \|c\|^2 \|w\|^2 \|U_c(a)\|.\end{aligned}$$

Here $\|U_v\| \leq \|v\|^2$, and $\|c\| \leq 1$. □

Lemma 3 (tail-product identity). *For c, x, y in a unital Jordan algebra,*

$$U_c x \circ U_c y = U_c(U_{x,y}(c^2)). \quad (7)$$

Proof. Apply the fundamental identity to the unit:

$$(U_c z)^2 = U_{U_c z}(\mathbf{1}) = U_c U_z U_c(\mathbf{1}) = U_c(U_z(c^2)).$$

Substitute $z = x + y$, subtract the corresponding formulas for x and y , and use

$$(r + s)^2 - r^2 - s^2 = 2r \circ s, \quad U_{x+y} - U_x - U_y = 2U_{x,y}.$$

Dividing by 2 gives (7). Thus the identity uses only the fundamental Jordan identity and is valid also in exceptional Jordan algebras. □

Combining Lemmas 2 and 3 with (2) gives the estimate that drives the proof: if $x, y \in M(\mathfrak{A})_{\text{sa}}$, $\|x\|, \|y\| \leq C$, and c is a self-adjoint contraction, then for $a \in \mathfrak{A}_+$,

$$\begin{aligned}\rho_a(U_c x \circ U_c y)^2 &= \|a \circ U_c(U_{x,y}(c^2))\|^2 \\ &\leq 9C^4 \|a\| \|U_c(a)\|.\end{aligned} \quad (8)$$

Main theorem

Theorem 4. *Let $\mathfrak{A}$ be a JB^* -algebra.*

(i) *The Jordan product on $M(\mathfrak{A})$ is separately J -strict continuous.*

(ii) *If $x_\lambda \rightarrow x$ and $y_\mu \rightarrow y$ J -strictly and both nets are norm bounded, then*

$$x_\lambda \circ y_\mu \longrightarrow x \circ y$$

J -strictly on the product directed set.

Consequently Questions Q1 and Q2 both have affirmative answers.

Proof. We first work in the self-adjoint part and test against $a \in \mathfrak{A}_+$.

Joint continuity at $(0, 0)$. Suppose $x_\lambda \rightarrow 0$ and $y_\mu \rightarrow 0$ J -strictly, with $\|x_\lambda\|, \|y_\mu\| \leq C$. Let (e_γ) be a positive contractive approximate unit and set $c_\gamma = \mathbf{1} - e_\gamma$. For a fixed γ , Lemma 1 gives

$$\|L_{e_\gamma} x_\lambda\| \longrightarrow 0, \quad \|L_{e_\gamma} y_\mu\| \longrightarrow 0.$$

Using $x_\lambda = L_e x_\lambda + R_e x_\lambda$, and suppressing the fixed subscript γ , write

$$x_\lambda \circ y_\mu = (L_e x_\lambda) \circ y_\mu + (R_e x_\lambda) \circ (L_e y_\mu) + (R_e x_\lambda) \circ (R_e y_\mu).$$

The first two terms tend to zero in norm on the product directed set:

$$\begin{aligned}\|(L_e x_\lambda) \circ y_\mu\| &\leq C \|L_e x_\lambda\|, \\ \|(R_e x_\lambda) \circ (L_e y_\mu)\| &\leq C \|L_e y_\mu\|.\end{aligned}$$

For the last term, (8) says

$$\rho_a((R_e x_\lambda) \circ (R_e y_\mu))^2 \leq 9C^4 \|a\| \|U_{1-e}(a)\|.$$

Given $\varepsilon > 0$, first choose γ so that the right-hand side is smaller than $\varepsilon^2/9$, using (1). With this $e = e_\gamma$ fixed, choose λ, μ late enough that the first two terms contribute less than $2\varepsilon/3$ after applying ρ_a . Hence $\rho_a(x_\lambda \circ y_\mu) \rightarrow 0$.

Separate continuity at zero. Fix $x = x^* \in M(\mathfrak{A})$, and let $z_\lambda = z_\lambda^* \rightarrow 0$ J-strictly with $\|z_\lambda\| \leq C$. Decompose with the same e, c :

$$x \circ z_\lambda = (L_e x) \circ z_\lambda + (R_e x) \circ (L_e z_\lambda) + (R_e x) \circ (R_e z_\lambda).$$

Because $L_e x \in \mathfrak{A}$,

$$\|(L_e x) \circ z_\lambda\| = \rho_{L_e x}(z_\lambda) \rightarrow 0.$$

The second term tends to zero in norm by Lemma 1. For the third, the proof of (8) gives

$$\rho_a((R_e x) \circ (R_e z_\lambda))^2 \leq 9\|x\|^2 C^2 \|a\| \|U_{1-e}(a)\|.$$

Choose e late in the approximate unit and then λ late. This proves $x \circ z_\lambda \rightarrow 0$ J-strictly.

All test elements and complex elements. Every self-adjoint $a \in \mathfrak{A}$ is a difference of two positive elements, and every element of $\mathfrak{A}$ is a complex linear combination of self-adjoint elements. Thus positive test elements determine J-strict convergence. Moreover

$$\rho_a(z^*) = \rho_{a^*}(z).$$

Consequently the real and imaginary parts of a bounded J-strict-null net are bounded J-strict-null self-adjoint nets. Expanding the Jordan product complex-bilinearly reduces both assertions just proved to the self-adjoint case.

Arbitrary limits. For bounded J-strictly convergent nets, the difference nets are bounded, and

$$\begin{aligned}x_\lambda \circ y_\mu - x \circ y &= (x_\lambda - x) \circ (y_\mu - y) \\ &\quad + x \circ (y_\mu - y) + (x_\lambda - x) \circ y.\end{aligned}$$

The first term tends to zero by joint continuity at $(0,0)$, and the last two by separate continuity. This proves (ii).

Finally, every J-strictly convergent net is J-strict bounded and hence norm bounded by the bounded-set theorem in [1]. The separate-continuity argument therefore applies to every convergent net, proving (i). $\square$

Verification notes

- The proof is internal to Jordan algebra theory. It does not embed $\mathfrak{A}$ into an associative algebra, so exceptional JB*-algebras are covered.

- As an independent guard against a special-only identity, the fundamental identity and (7) were evaluated in 40 exact random trials in the exceptional 27-dimensional real Albert algebra. The script is included as `code/check_albert_identities.py`. Computation is only a sanity check; Lemma 3 is the proof.
- The source crop is reproducible with `code/make_open_questions_crop.py`.
- The bounded novelty search cannot exclude an unindexed or differently phrased answer. Novelty confidence is therefore moderate.

References

- [1] F. J. Fernandez-Polo, J. J. Garcés, L. Li, and A. M. Peralta, *On the strict topology of the multipliers of a JB^* -algebra*, Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Mat. RACSAM **117** (2023), Paper No. 146; arXiv:2210.13353.
- [2] H. Hanche-Olsen and E. Størmer, *Jordan Operator Algebras*, Pitman, London, 1984.